

Jewel G. Padilla

Raleigh, NC | jgpadill@ncsu.edu | [LinkedIn](#) | [Google Scholar](#) | [Portfolio](#)

Education

North Carolina State University, Raleigh, NC

Ph.D. in Human Factors and Applied Cognition

Advisor: Dr. Colleen Patton

Exp. May 2029

North Carolina State University, Raleigh, NC

Bachelor of Arts in Psychology

GPA: 4.0

Dean's List: Spring 2023, Fall 2023, Spring 2024

December 2024

Research Interests

My primary research focuses on examining the interactions and underlying decisions of individuals utilizing automation. With the increase in automation, artificial intelligence, and machine learning, I look to understand how different factors impact human-automation usage and how those interactions can be optimized to support real-world challenges.

Keywords: human factors, automation, perception, human-automation teaming, expertise, risk, workload

Publications

- **Padilla, J. G.,** Patton, C. E., (2025). Perceptions of Difficulty In Automation-Supported Task. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. 69(1).
- Perez-Cabarcas, M., Yakar, U., Hsu, I., Josyula, K., Lee, R.A., Harris, C., Destin, A., **Padilla, J. G.,** Addison, I., Mao, M., Lin, X., Mohsenin, T., McDowell, K., [In Press]. COVERT-AI: A Trustworthy Multimodal System for Adaptive Real-Time Cognitive Support. *HCHI 2026 Conference Proceedings*.

Manuscripts in Review

- **Padilla, J. G.,** Patton, C. E. (Under Review). The Impact of Time Pressure on Perceptions of Difficulty in an Automated-Supported Task. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. 70(1).
- **Padilla, J. G.,** Patton, C. E. (Under Review). Knowing When to Use Automation: Task-specific expertise and GPS engagement in a navigational task. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. 70(1).

Collaborative and Laboratory Research Projects

Research and Data Science Intern

June 2026 – Current

Summer Conference on Applied Data Science (SCADS), Laboratory for Analytic Sciences, NC State University

- Designed a within-subjects 2x2 factorial protocol to evaluate an agentic AI sensemaking harness against manual multi-system workflows, using block randomization, AB/BA counterbalancing, and matched parallel-form scenario packets.
- Formalized hypotheses spanning efficiency, performance under complexity, and traceability, operationalizing outcomes as separable measures with phase-level timing instrumentation.
- Built a multi-method measurement battery integrating validated instruments (Raw NASA-TLX, Trust in Automation, SUS) and a seeded-error paradigm to quantify automation bias via per-participant catch-rate scoring.
- Specified human factors evaluation for a Claude Agent SDK orchestrator-worker harness with per-store MCP connectors, defining blinded inter-rater rubrics and effort-reallocation analyses across graph, relational, and unstructured data stores.
- Ran timing pilots and authored an inter-rater scoring manual to validate session feasibility and lock pre-specification of sample size, exclusion rules, and multiplicity correction.

Graduate Researcher

June 2025 – Current

CHAT Lab, Human Factors and Applied Cognition Program, NC State University

- Design and conduct experiments on human-automation interaction and risk perception, including paradigm development, stimulus generation, and IRB protocol preparation and submission.
- Apply parametric and non-parametric statistical tests (t-tests, ANOVA, regression, Wilcoxon) with assumption checking and effect-size estimation to evaluate behavioral patterns.
- Build analysis and visualization workflows in RStudio to translate experimental data into structured metrics and present findings to diverse audiences.
- Authored and presented a first-authored paper at the Human Factors and Ergonomics Society (HFES) Annual Meeting.

UX Researcher & Human Factors Consultant**September 2025 – January 2026***Designcorps & NC State University, Raleigh, NC*

- Led an interdisciplinary team as part of a graduate level human factors course capstone in collaboration with the Design Department at NC State University.
- Conducted UX research using heuristic evaluation, task analysis, and experimental testing to evaluate prototype tactile warning indicator surfaces (TWSIs) designed for users with visual impairments.
- Presented findings to stakeholders showing significant usability and design improvements.

Human Cognitive Enhancement Research Intern**June 2025 – August 2025***DEVCOM Army Research Lab (ARL), Humans in Complex Systems (HCxS), Aberdeen Proving Ground, MD*

- Cross-functionality worked with engineers, computer scientist, and psychologist on a research project studying the enhancement of mental models by developing a multi-modal retrieval-augmented generation (RAG) cognitive assistant that helps users navigate information-dense environments.
- Designed and conducted formative user evaluations of the human-AI interaction layer, measuring decision accuracy, response latency, and reliance on system confidence cues to inform interface changes to confidence signaling, information hierarchy, and intervention timing.
- Transformed raw interaction logs and experimental data into structured, analysis-ready datasets, treating override rates, error-correction behavior, and response-time distributions as evaluative signals communicated to developers as quantitative metrics.
- Translated human-factors principles into actionable system requirements during the platform's transition from a monolithic desktop application to a containerized microservice architecture with a React-based front end.
- Communicated findings through visualizations, technical summaries, presentations, posters, and technical reports for a multidisciplinary team of ML engineers, systems architects, data engineers, and government stakeholders.

CHAT Lab Manager,**January 2025 – May 2025***Human Factors and Applied Cognition, NC State University, Raleigh, NC*

- Analyzed collected data with statistical analysis in Excel and R Studio and prepared comprehensive reports of findings.
- Assisted in writing extended abstracts and documents to submit for conferences.
- Trained and mentored research assistants on research and experiment protocols.
- Interviewed and hired potential research assistants.

CHAT Lab Undergraduate Student Researcher,**January 2024 - December 2024***Human Factors and Applied Cognition, NC State University, Raleigh, NC*

- Collaborated with Dr. Colleen Patton and graduate students; maintained weekly logs, led article discussions, and provided feedback on pilot studies.
- Supported project development by reviewing literature and gaining hands-on experience in independent research on human-automation interaction.

LEMON Lab Research Study Assistant,**May 2024 – June 2024***Czech Academy of Sciences, Prague, Czech Republic*

- Collaborated with researchers studying phonetic acquisition by conducting a comprehensive literature review on infant babbling.
- Learned and assisted different methodologies for measuring early speech, including EEG and acoustic analysis.
- Observed EEG administration sessions and contributed to ongoing studies by supporting data collection and engaging in discussions on experimental design.

ERPAS Undergraduate Student Researcher,**August 2023 - December 2023***School Psychology, NC State University, Raleigh, NC*

- Reviewed scholarly articles and chapters on educational research, educational theory, social justice, and community-engaged scholarship.
- Developed potential materials for research and service projects.

Independent Research Projects**Perceptions of Difficulty in Automation-Supported Task****September 2024-Current***CHAT Lab, Human Factors and Applied Cognition Program, NC State University*

- Performed quantitative statistical analysis through RStudio.
- Effectively communicated findings to a broad audience through a published paper.
- Delivered a lecture communicating research findings to a diverse audience.
- Findings highlighted the role of design in improving employee satisfaction, trust, and accessibility.

Examining Risk and Expertise in Automation Use Through GPS Systems**August 2024 - Current***CHAT Lab, Human Factors and Applied Cognition Program, NC State University*

- Conducted a literature review on expertise, risk, and automated navigation to inform the development of a focused research question and hypothesis.
- Designed an experimental paradigm, including tasks, maps, and procedures, to test how expertise influences risk perception in automation.
- Prepared IRB materials (consent, debrief, and exemption forms) and authored a detailed methods section outlining participants and study design.
- Currently conducting data collection for a human-subjects and preparing for statistical analyses.

Perception of Risk: Examining the Influence on Automated Vehicle Usage**April 2024 – August 2024***CHAT Lab, Human Factors and Applied Cognition Program, NC State University*

- Investigated the relationship between situational risk and trust in automated vehicles.
- Created and administered a two-phase survey to assess participants' perceptions of risk and preference in automated versus manual driving.
- Applied statistical methods, including t-tests, ANOVA, and regression analysis to confirm the significance of findings.
- Utilized data visualization tools, including RStudio, to perform statistical analysis and present findings at local conferences.

Teaching**Instructor***PSY 420: Cognitive Processes, North Carolina State University***June 2026 – Current**

- Design and teach an undergraduate cognitive psychology course covering attention, memory, perception, and decision-making.
- Build the full curriculum, including lectures, assignments, and assessments aligned to course learning objectives.
- Lead class sessions and mentor students through office hours and individualized feedback.
- Evaluate student performance and assign grades across the term.

Grants and Fellowships**Graduate Research Assistantship (GA)***North Carolina State University, Department of Psychology, Human Factors and Applied Cognition Program*

- Competitive assistantship awarded to support select Ph.D. students.

University Graduate Fellowship for 2025-26 (\$4,000); North Carolina State University, Office of the Provost

- Prestigious funding award to support incoming students' efforts towards a Ph.D. in Psychology.

2024 Provost Experience Program (\$1,000); North Carolina State University, Office of Undergraduate Research

- Funds awarded to undergraduate students to provide compensation for necessary contributions to ongoing research in a faculty laboratory.

Fall 2024 OUR Travel Award; North Carolina State University, Office of Undergraduate Research

- Funds awarded to undergraduate students to formally present their research at a conference.

Summer 2024 OUR Research Assistant Grant; North Carolina State University, Office of Undergraduate Research

- Competitive funding awarded to undergraduate students to support independent research projects throughout a semester.

Leadership and Service**Vice President***Human Factors and Ergonomics Society (HFES), NC State University Student Chapter***May 2026 - Current**

- Support the chapter president in planning professional development events, including guest speakers and workshops.
- Coordinate chapter meetings and member communications to sustain engagement across the student body.
- Assist in recruiting and onboarding new members to grow chapter participation.
- Represent the chapter in collaboration with the Human Factors and Applied Cognition program and national HFES.

Student Reviewer*Human Factors and Ergonomics Society (HFES) Annual Meeting***February 2026 - Current**

- Evaluated submissions for the HFES Annual Meeting across General Sessions and Human-AI-Robot Teaming (HART).
- Provided structured feedback to authors on methodology, analysis, and clarity to support publication quality.

Research Mentor**August 2025 - Current***CHAT Lab, Human Factors and Applied Cognition Program, NC State University*

- Mentored an undergraduate researcher supporting a human-automation interaction project, guiding their development of an independent subproject.
- Advised on study design, analysis, and presentation, culminating in their presentation at the NC State Undergraduate Research & Creativity Symposium.
- Wrote letters of recommendation supporting their academic and professional applications.

Presentations and Posters

- **Padilla, J. G.,** Patton, C. E., (2026). *Who Uses Automation? Effects of Spatial Ability, Confidence, and GPS Use*. [Poster, accepted]. American Psychological Association (APA) Annual Convention, Washington, D.C.
- Santana, O., **Padilla, J. G.,** Patton, C. E. (2026). *The Effects of Confidence and Spatial Knowledge on GPS Usage*. [Poster]. Presented at the NC State Undergraduate Research & Creativity Symposium. Raleigh, NC.
- **Padilla, J. G.,** Patton, C. E., (2025). *Perceptions of Difficulty In Automation-Supported Task*. [Lecture]. Presented at the Human Factors and Ergonomics Society Annual Meeting. Chicago, IL.
- **Padilla, J. G.,** (2025). *COVERT-AI for Future C2 and Army Use*. [Poster]. Presented at the Army Research Lab (ARL) Summer Research Develop Program Symposium. Aberdeen Proving Ground, MD.
- Perez-Cabarcas, M., Destin, A., Harris, C., **Padilla, J. G.,** Lee, R. A., Hsu, I., Yakar, U., Josyula, K., (2025). *COVERT-AI for Mental Model Formation*. [Talk]. Presented at the Army Research Lab (ARL) Humans in Complex Systems (HCxS) Symposium. Aberdeen Proving Ground, MD.
- Perez-Cabarcas, M., Destin, A., Harris, C., **Padilla, J. G.,** Lee, R. A., Hsu, I., Yakar, U., Josyula, K., (2025). *COVERT-AI for Mental Model Formation and Multi-RAG Use*. [Talk] Presented at the Department of Defense (DoD) Basic Research Conference. College Park, MD.
- **Padilla, J.,** Patton, C. E., (2024). *Perception of Risk: Examining the Influence on Automated Vehicle Usage*. [Poster]. Presented at the NCSU Summer 2024 Undergraduate Research & Creativity Symposium. Raleigh, NC.
- **Padilla, J.,** Patton, C. E., (2024). *Impact of Risk Perception on Automated Vehicle Utilization in Driving Scenarios*. [Poster]. Presented at the National Research Conference at Penn. Philadelphia, PA.

Professional Memberships

- Student Member, Human Factors and Ergonomics Society (HFES)
- Student Member, American Psychological Association (APA)

Certifications

- CITI Program, Responsible Conduct of Research – [2026]
- CITI Program, Human Subjects Research – [2024]

Relevant Skills

- **Data Analysis & Statistics Skills:** R-Studio, Microsoft Excel, Qualtrics, SPSS, SAS, Python. Mixed-effects/multilevel models, signal detection analysis, non-parametric tests (Wilcoxon, Mann-Whitney), effect size estimation, power analysis.
- **Usability Testing & User Feedback:** Conduct usability tests and analyze user feedback to identify issues and behavioral patterns driving design improvements.
- **User Research & Experimental Design:** Conduct user interviews and surveys, design and run experimental studies to inform UX improvements, task and heuristic analysis, cognitive walkthroughs.
- **Experimental & Research Methods:** IRB protocol preparation and submission, experimental paradigm design, stimulus generation, pilot and engineering testing, NASA-TLX and workload measurement, survey design
- **Research Skills:** Understanding of research methods, data and statistical analysis, literature review.
- **Domain Knowledge:** Human-automation interaction, trust and reliance, automation bias, situation awareness and workload, retrieval-augmented generation (RAG) systems, AI systems, machine learning.
- **Technical Tools:** PsychoPy, Python, GitHub, Slack, Microsoft Office Suite, Canva.